

Cyclone Core MVP Roadmap

M0 → M3 · 12 weeks

Walking Skeleton quick validation → ADAS domain-controller SiL loop into CI · C++17 real-time foundation · weekly deliverables, one decision every three weeks

M0 Walking Skeleton W1–2

- C++17 foundation + TaskThread / fmt wrappers
- Scheduler v0 + Time Master abstraction
- FMU runner (motor demo model)
- Lua bindings + minimal MAPort subset
- JSON report, first end-to-end pass

Exit FMU → script → report chain runs end-to-end

M1 Real-Time Kernel W3–5

- Time Master dual-mode clock (real-time + PTP reservation)
- Scheduler v1: jitter < 500µs on target
- FlatBuffers shared-memory zero-copy message bus
- SocketCAN / UDP adapters (IBusAdapter)

Exit Unified timebase + bus I/O working

M2 Data Loop W6–8

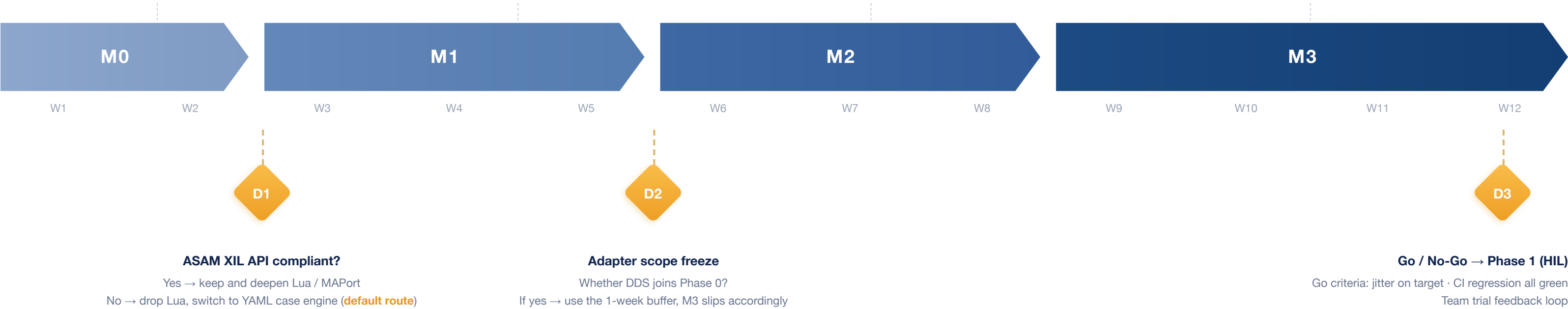
- MF4 / rosbag replay engine (time-aligned, rate-scaled)
- MDF4 measurement: ring buffer + side-channel dump
- Docker SUT lifecycle management (start / stop / inject)
- OSI target-level injection interface

Exit Replay → perception feed → measurement loop closed

M3 Automation & Acceptance W9–12

- YAML case engine + measurement-flow real-time verdicts
- JUnit XML report + CLI / REST API
- Jenkins / GitLab CI integration
- AEB scenario end-to-end regression validation (factory reference cases)

Exit CI regression all green, team daily-use ready



Three standing disciplines (C++17 debt guard)

Threads only via TaskThread wrappers · formatting only via fmt · no allocation on the RT path (PMR pools)
A future C++20 upgrade swaps the wrappers, not the business code



Phase 1 preview: HIL

PREEMPT_RT real-time → PTP rig time sync → residual bus simulation
OpenSCENARIO scenario engine (integrating the esmini player)